FLATNESS OF THICKENINGS MORPHISMS DESCENDS TO THE CLOSED FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 663-684 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 686-708, the complete author proof (21 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 43 of queue_stacks-stacks-more-morphisms.json (pb=969, stmt 663-684, proof 686-708).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned. Named omission: the pointwise flatness follows 'in the same manner' - identical reduction, mechanism named.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0CGW: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 708–708. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Consider a commutative diagram*

$$\begin{array}{ccc} (\mathcal{X} \subset \mathcal{X}') & \xrightarrow[(f,f')]{} & (\mathcal{Y} \subset \mathcal{Y}') \\ & \searrow \quad \swarrow & \\ & (\mathcal{B} \subset \mathcal{B}') & \end{array}$$

of thickenings of algebraic stacks. Assume

(1) $\mathcal{Y}' \to \mathcal{B}'$ *is locally of finite type,*
(2) $\mathcal{X}' \to \mathcal{B}'$ *is flat and locally of finite presentation,*
(3) f *is flat, and*

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(4) $\mathcal{X} = \mathcal{B} \times_{\mathcal{B}'} \mathcal{X}'$ *and* $\mathcal{Y} = \mathcal{B} \times_{\mathcal{B}'} \mathcal{Y}'$.

Then f' is flat and for all $y' \in |\mathcal{Y}'|$ in the image of $|f'|$ the morphism $\mathcal{Y}' \to \mathcal{B}'$ is flat at y'.

Proof. Choose an algebraic space U' and a surjective smooth morphism $U' \to \mathcal{B}'$. Choose an algebraic space V' and a surjective smooth morphism $V' \to U' \times_{\mathcal{B}'} \mathcal{Y}'$. Choose an algebraic space W' and a surjective smooth morphism $W' \to V' \times_{\mathcal{Y}'} \mathcal{X}'$. Let U, V, W be the base change of U', V', W' by $\mathcal{B} \to \mathcal{B}'$. Then flatness of f' is equivalent to flatness of $W' \to V'$ and we are given that $W \to V$ is flat. Hence we may apply the lemma in the case of algebraic spaces to the diagram

$$\begin{array}{ccc} (W \subset W') & \longrightarrow & (V \subset V') \\ & \searrow \quad \swarrow & \\ & (U \subset U') & \end{array}$$

of thickenings of algebraic spaces. See More on Morphisms of Spaces, Lemma 0CGW. The statement about flatness of $\mathcal{Y}'/\mathcal{B}'$ at points in the image of $|f'|$ follows in the same manner. □